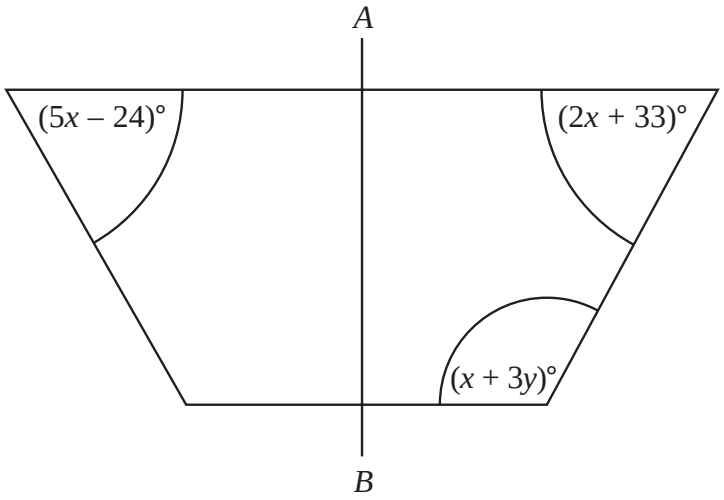
**7** The diagram shows a trapezium.
*AB* is the line of symmetry of the trapezium.




Work out the value of $x$ and the value of $y$.

Symmetry: $5x - 24 = 2x + 33$

**1 mark**
Equal symmetric top angles.

$3x = 57$, so $x = 19$

**2 marks total**
Collect x terms; solve x = 19.

Top-right angle = $2(19) + 33 = 71°$

$71 + 19 + 3y = 180$

$3y = 90$, so $y = 30$

**2 marks total**
Co-interior angles between the parallel bases sum to180°.

$x =$ 19

$y =$ 30

**(Total for Question 7 is 5 marks)**

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